

Alex R. Atrio, PhD

Senior Scientist | LLM Specialist | AI Project Lead



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🗣 Languages: English, French, Spanish, Catalan, Italian

Summary

- PhD in Natural Language Processing (EPFL)
- Leading €1M ESA project (EVE): open-source LLM for Earth Observation
- 6+ years of experience in NLP/AI research and applied projects
- Strong specialization in LLMs: pretraining, fine-tuning, alignment, and RAG
- **Project management:** budgeting, mentoring team of 5, stakeholder coordination
- **Prolific scientific publishing:** EACL, EAMT, etc.
- Track record in **grant success:** EuroHPC, Runpod, multiple EU/ESA grants
- **Focus:** domain adaptation, low-resource NLP, and regularization methods

Skills

Focus LLMs · Low-Resource NLP · Machine Translation · Pre-training · Fine-Tuning · RAG
Strategy Scientific Writing · Grant Development · Project Management · Agile
Toolkit PyTorch · Hugging Face · Langchain · Asana · Git · Google Workspace
Coding Python, Prolog, $\LaTeX$

PROFESSIONAL EXPERIENCE

Pi School

Rome, Italy

Project lead – Senior Deep Learning Scientist

2024 - Present

- **Project lead** for EVE: Managing an 18-month, €1M project to develop an open-source LLM for Earth Observation for Φ -lab, the innovation lab at the European Space Agency:
 - * **Technical responsibilities:** overall technical direction, decision-making, research on SOTA.
 - * **LLM key areas:** data processing, continued pre-training, instruction fine-tuning, alignment, retrieval-augmented generation.
 - * **Management responsibilities:** mentoring and guiding 5 engineers, stakeholders discussions, budget oversight.
- Contributor to multiple **winning grant proposals**, including Videoloc (FBK €150k), SatcomLLM (ESA €400k)
- **Project lead** for Videoloc: Managing a 12-month, €150k project to optimize automatic subtitle generation based on professional post-editing data for Fondazione Bruno Kessler (FBK), a research institute in Italy focused on AI.

HEIG-VD

Lausanne, Switzerland

Teaching Assistant

2019 - 2023

- Final undergraduate NLP courses in French. Guiding students and reviewing laboratory work.

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL)

Lausanne, Switzerland

PhD in Natural Language Processing

2019 - 2023

- Thesis: [Regularization Techniques for Low-Resource Machine Translation](#).
- Supervisor: [Andrei Popescu-Belis](#).
- I also contributed to applied projects with external partners:
 - * Armasuisse - [UNISUB](#): end-to-end MT optimization through adaptive scheduling of training tasks.
 - * Armasuisse - [FamilyMT](#): design systems for closely related source languages.
 - * SNSF Agora - [CRPO](#): development deployment of an interactive poetry generation system.

University Pompeu Fabra & University of Barcelona

Barcelona, Spain

Masters in Cognitive Science

2017 - 2019

- Thesis: [MT Reordering as Preprocessing for Cross-Lingual Sentiment Analysis](#).
- Supervisors: [Toni Badia](#) & [Jeremy Barnes](#).

University of Barcelona

Barcelona, Spain

Bachelor in Philosophy

2012 - 2017

RESEARCH

I have published across several areas in NLP. For the full list, please visit my [Google Scholar profile](#).

- [Atrio, À.R., et al. \(2023\). \[A Simplified Training Pipeline for Low-Resource and Unsupervised MT \\(LORES 23\\)\]\(#\)](#)
- [Popescu-Belis, A., Atrio, À.R., et al. \(2023\). \[GPoeT: a Language Model Trained for Rhyme Generation on Synthetic Data \\(LaTeCH-EACL 23\\)\]\(#\)](#)
- [Atrio, À.R., Popescu-Belis, A. \(2022\). \[On the Interaction of Regularization Factors in Low-Resource NMT \\(EAMT 22\\)\]\(#\)](#)
- [Atrio, À.R., Popescu-Belis, A. \(2021\). \[Small Batch Sizes Improve Training of Low-Resource NMT \\(ICON 21\\)\]\(#\)](#)

Additionally, I have served in the Scientific Committee of the [NASA/ESA workshop on AI FM models](#) and [ESA's LPS25](#).

AWARDS

EuroHPC

2025

- Principal Investigator. [Won Grant](#): 3,500 node hours (~€31,000) on the MareNostrum5 supercomputer.

Runpod

2024

- Principal Investigator. [Won Startup and Research Grant](#): €3,000 in compute credits.

SEPLN

2019

- First author. Best Paper Award for [Master's thesis](#) - on word order in multilingual sentiment analysis.

INTERESTS

- Projects with state-of-the-art technology.
- Development prospects and responsibility.
- Open, friendly and agile working environment.
- Balance between application and scientific research.

REFERENCES

Andrei Popescu-Belis

- Professor of Computer Science, EPFL & HEIG-VD
- andrei DOT popescu DASH belis AT heig-vd.ch
- iict-space.heig-vd.ch/apu/